

Tudor Dolineaschi has completed the 2024 Stanford Pre-Collegiate Summer Institutes course in

Introduction to Machine Learning

from June 17th to June 28th, online with Stanford University. Tudor's admission into the Stanford Pre-Collegiate Summer Institutes and successful completion of the program demonstrate a high level of academic ability and intellectual curiosity.

Course Description:

Companies worldwide are utilizing sophisticated machine learning techniques to learn from data, identify patterns, and accurately predict industry-specific outcomes. This course gave students an overview of the machine learning process and introduced them to advanced techniques commonly used to extract insights from large datasets. Using the Python programming language, students employed several steps to preprocess and visualize structured data; trained, tested, and tuned machine learning models; and evaluated algorithm performance to select a final model for presentation.

Student Evaluation (from Instructor):

In this course, Tudor was taught the key differences between supervised and unsupervised machine learning and how to distinguish between classification and regression tasks. He received hands-on training using the Python programming language to analyze datasets, construct univariate/bivariate plots and correlation matrices, and employ supervised learning methods and techniques for binary classification. For the course project, Tudor searched through a machine learning data repository to select an open dataset that met the criteria for classification tasks. Tudor used several binary classification algorithms and ensemble methods on their dataset to train, test, and evaluate machine learning models with metrics that included accuracy, recall, precision, and F1 scores. Tudor also designed a research poster highlighting his work and subsequently presented his poster to the class as a rapid-fire (five minute) presentation.

Overall, Tudor performed very well in the course. Throughout the two weeks, Tudor demonstrated consistent effort in learning advanced topics in artificial intelligence and machine learning. He actively participated in our sessions by answering challenging questions when called upon in class, and he attended office hours to ask insightful questions that informed his machine learning research project. For his final project, Tudor trained, tested, and evaluated six machine learning models that predicted pet adoption, using the decision tree, random forest, and the bootstrap aggregating (bagging) classifier as binary algorithms/ensemble methods (as well as tuned versions of these techniques). Tudor was able to accurately interpret evaluation metrics for each of his models and identified the features that served as the strongest predictors of his outcome. Tudor's research poster and final presentation were well received by his peers.

It was a pleasure to have Tudor in my Machine Learning course this summer. I look forward to seeing Tudor participate in other AI courses at Stanford in the future.



Dr. Jamie Fairclough